# Hospital Information Management System (HIMS)

End-to-End Inpatient Operational, Clinical, & Financial Workflow Blueprint

## 1. Patient Intake & Identity Creation

**IP System Generation:** Upon formal admission, the core service generates a unique, immutable Inpatient Code matching the format: **IP-YYYY-NNNNN** (e.g., *IP-2026-04921*). This serves as the primary foreign key across clinical, operational, and financial ledgers.

**Triage & Clinical Routing:**

- **Critical (ICU):** Direct transfer to Intensive Care; triggers immediate high-intensity nurse allocation and specific monitoring matrices.
- **Non-Critical (Ward):** Routed to General, Semi-Private, or Private Wards based on clinical advisory and availability.

**Financial Triaging & Triage Channels:**

- **Insurance Route:** System triggers an automated Pre-Authorization API payload to the payer clearinghouse. Records approved caps, deductible limits, and sets the dynamic co-pay percentage variable inside the patient profile.
- **Cash/Self-Pay Route:** Flagged immediately as a Self-Pay account. Calculates and requires a minimum upfront room deposit based on standard tariff tables prior to full ward activation.

## 2. Clinical Tracking & Attendance Logging

**Doctor Round Metrics:** Every physical round triggers a relational log requiring: Doctor ID, timestamped entry/exit, specific clinical notes, and structural orders tied to CPT (Current Procedural Terminology) codes.

**Nurse Attendance & Shift Logs:** Continuous updates are managed via a relational table tracking shift assignment, vital check intervals, and real-time updating of the Medication Administration Record (MAR).

**Intra-Hospital Transfers:** When a patient shifts from ICU to a Ward, the system performs an atomic database update. Room billing calculations transition exactly at the logged checkout/check-in timestamp to prevent dual-billing anomalies.

## 3. Live Cash Flow & Charge Capturing

The system maintains a running ledger that captures every transaction event dynamically:

| Charge Category | Triggering Mechanism | Ledger Impact |
|---|---|---|

| | | |
|---|---|---|
| Room/ICU Fees | Automated daily/hourly cron job matching timestamp location. | Debited to running account balance. |
| Diagnostics/Labs | Barcode verification scan upon lab sample processing arrival. | Real-time debit post-analysis execution. |
| Consumables/Meds | Dispensation validation scans via ward inventory modules. | Immediate decrement from inventory & debit to patient. |

## 4. The 4-Tier Discharge & Clearance Gateway

To ensure full operational compliance and mitigate revenue leakage, a sequential digital clearance gateway enforces a hard dependency structure:

**Gate 1: Doctor Discharge Seal**
The primary consulting practitioner triggers the medical discharge order. System autogenerates the Clinical Discharge Summary and compiles the take-home medical prescription schema.

**Gate 2: Pharmacy Clearance Seal**
The central pharmacy reconciles unused ward medications, dispenses the take-home prescription, updates inventory, and places the *Pharmacy Clearance Seal*.

**Gate 3: Nursing Bill Review Seal**
The head floor nurse reviews the physical file against system records, audits consumables utilized, confirms zero active vitals tracking, and sets the *Nursing Clearance Seal*.

**Gate 4: Administration Audit & Final Release Seal**
The billing desk initiates automated final balance processing:
**Formula:** Total Gross Charges - Insurance Covered Limit - Initial Cash Deposits = *Net Outstanding Due*.
• If Net Due > 0, system locks workflow until physical cash/card payment confirmation is pushed.
• Once balanced, the *Final Admin Discharge Seal* releases, exporting the patient summary.

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